AKUL VARSHNEY

Senior Backend | AI Enabled Full-Stack Engineer | Node.js · React · AWS · Agentic / Generativ AI · LangChain · FastAPI

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PROFESSIONAL SUMMARY

Backend-first full-stack engineer with 3+ years of building and scaling distributed, multi-tenant systems in production. Led end-to-end architecture of an enterprise food supply chain platform processing 100K+ daily transactions across 300+ trains. Early adopter of production-grade GenAI — shipped invoice extraction and RFQ copilot features using the Anthropic SDK, and currently engineering multi-agent AI pipelines with LangGraph/LangChain and Azure OpenAI at EXL. Proven track record of 40–75% performance and cost gains through system re-architecture. Seeking a senior engineering role to own complex backend systems and drive AI-enabled product development.

KEY ACHIEVEMENTS

- Architected and scaled a Greenfield food supply chain platform to 100K+ daily transactions across 300+ trains with zero data loss using BullMQ async pipelines and SAP ERP integration.
- Reduced API response time by 75% (2s → 500ms) and cut infrastructure costs ~75% by Dockerizing a previously outsourced fleet management system.
- Achieved 40% API latency improvement on the IRCTC vendor platform by profiling and restructuring MongoDB queries and indexing strategies independently.
- Shipped production GenAI features — AI-driven invoice extraction and an RFQ copilot with guardrails — using the Anthropic SDK in an enterprise vendor management portal.
- Engineered multi-agent AI pipelines using LangGraph and Azure OpenAI (GPT-4o) for automated claims document processing with Human-in-the-Loop review workflows at EXL.
- Delivered a full transport management system for Ethiopian Electric Power covering 500+ vehicles: bookings, dispatch, fuel management, and real-time GPS tracking.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C++

Backend: Node.js, Express.js, FastAPI, Flask, REST & GraphQL APIs, WebSockets, Microservices, BullMQ, LangChain, LangGraph

Frontend: React.js, Next.js, React Native, Redux, Tailwind CSS, Material UI, Ant Design, Framer Motion

Databases: PostgreSQL, MongoDB, MySQL, Redis, ChromaDB, Prisma ORM, Alembic — optimization, indexing, schema design

Cloud & DevOps: AWS (EC2, S3, Lambda, CloudWatch), Azure (OpenAI, AI Document Intelligence), GCP, Docker, GitHub Actions, CI/CD

Architecture: Distributed systems, Event-driven architecture, Message queues, Multi-agent pipelines, ACID, System design

Auth & Security: OAuth 2.0, JWT, RBAC, Aadhaar eSign, compliance workflows

AI & GenAI: Anthropic SDK, Azure OpenAI (GPT-4o), LangGraph, LangChain, Prompt engineering, Agentic AI, AI workflow automation

Testing & Tools: Unit & integration testing, TDD, Agile/Scrum, SAP ERP integration, Twilio WhatsApp API

PROFESSIONAL EXPERIENCE

Consultant 2 — **EXL**, Gurgaon **Jul 2026 – Present**

AI Claims Intelligence Platform

- Engineered a sophisticated multi-agent pipeline using LangGraph and LangChain to automate the classification and extraction of structured data from heterogeneous claim documents (PDFs, images).
- Built robust, high-performance RESTful APIs using FastAPI to orchestrate AI workflows, manage system state, and seamlessly handle document uploads.
- Integrated Azure OpenAI (GPT-4o) for complex reasoning tasks (anomaly detection, narrative summarization, and policy compliance recommendations) and Azure AI Document Intelligence for OCR.
- Designed and implemented relational and vector database schemas using PostgreSQL, Alembic, and ChromaDB to store extracted JSON data, policy embeddings for semantic search, and an immutable audit log.
- Developed automated cross-document validation logic to flag data inconsistencies (e.g., date conflicts, cost outliers), seamlessly routing exceptions to a Human-in-the-Loop (HITL) review queue.

Software Engineer — **Orane Consulting Pvt. Ltd.**, Noida **Mar 2023 – Jun 2026**

AI Multi-Agent Orchestration Platform

- Led architecture of an AI-driven multi-agent orchestration framework inside the vendor management portal, transforming it from a click-based interface into a conversational, NLP-driven workflow engine.
- Integrated a conversational chatbot assistant enabling seamless natural language interaction with complex backend workflows, improving task execution speed and reducing user onboarding friction.

Lead Backend Engineer — National Food Supply Chain (RK Group / IRCTC Vendor)

- Architected a Greenfield, multi-tenant food supply chain platform handling 100K+ daily transactions across 300+ trains; owned full stack selection (Node.js, Express, MongoDB, Redis, BullMQ, AWS S3) within a 4–6-member team.
- Engineered fault-tolerant async pipelines integrating BullMQ and SAP ERP, ensuring zero data loss and system resilience during peak load windows.
- Introduced Redis caching and event-driven schedulers to reduce database load and stabilize high-concurrency performance.
- Independently profiled and optimized MongoDB queries and indexes, cutting API latency by 40%.

Full-Stack Engineer — Enterprise Vendor Platforms (BlackBerry, BIBA, Avaada)

- Built and deployed scalable MERN-stack platforms using Dockerized microservices on AWS EC2, with CloudWatch monitoring and GitHub Actions CI/CD.
- Implemented Aadhaar eSign, eStamp, IFSC, and MCA validations, streamlining compliant vendor onboarding workflows for global clients.
- Developed a WhatsApp notification engine via Twilio with webhook-based delivery and automated retries.
- Resolved critical API latency (40% improvement) by migrating heavy DB operations server-side and adding Redis caching.
- Built production GenAI features — AI-driven invoice extraction and RFQ copilot — using the Anthropic SDK with grounded responses and output guardrails.

Full-Stack Engineer — Fleet Management System (Ethiopian Electric Power)

- Delivered an end-to-end transport management system for 500+ vehicles covering bookings, approvals, dispatch, and fuel management.
- Implemented real-time bus tracking via Google Maps and GCP APIs; automated vehicle allocation, reducing manual dispatch errors significantly.
- Enforced RBAC with JWT across employee, manager, and fleet manager roles with dedicated workflow UIs.

Backend Engineer — Real-Time Reverse Auction Engine

- Built a low-latency bidding engine using WebSockets (Socket.io) with Pub/Sub architecture for real-time concurrent auctions.
- Ensured ACID compliance via concurrency controls and validations; added unit tests covering critical bid-state scenarios.

PERSONAL PROJECTS

Clinic ERP & Multi-Tenant SaaS Platform — *React · Node.js · Prisma · PostgreSQL* **Jun 2024 – Present**

- Built a multi-tenant SaaS platform with dynamic RBAC, enabling runtime role configuration via JWT-secured middleware.
- Developed an appointment engine with real-time conflict detection, availability checks, and automated rescheduling.
- Implemented a financial module covering invoicing, receipts, and a custom PDF engine (standard + thermal POS formats).
- Created real-time KPI and billing analytics dashboards using Recharts via REST APIs.

Real-Time Interview Platform — *React · Node.js · Flask · WebSockets · Python* **2023**

- Built a live interview platform with collaborative whiteboard, in-browser IDE (Monaco Editor), and real-time chat.
- Deployed a CNN-based facial emotion recognition model via Flask with real-time video streaming over WebSockets.
- Automated code evaluation via GitHub integration and test pipelines, eliminating manual assessment overhead.

EDUCATION

B.Tech, Information Technology — Amity University, Noida 2019 – 2023 | CGPA: 8.03 / 10

CERTIFICATIONS & RECOGNITION

- AI-First Product Management — Airtribe (in progress)
- AWS Academy Graduate — Cloud Foundations (Amazon Web Services)
- CCNAv7: Enterprise Networking, Security & Automation (Cisco)
- AI Coder: Vibe Coder to Agentic Engineer — Udemy
- Achiever's Award for Knowledge Creation & Research Contribution
- Published Research: Emerald (MSME Resilience), Springer (AI & Cloud Software), IEEE (Face Mask Detection & e-Challan)

CORE COMPETENCIES & SOFT SKILLS

Distributed Systems Architecture · Event-Driven Architecture · Multi-Agent AI Pipelines · Microservices · System Design & Scalability · API Design (REST/GraphQL) · Cloud Architecture (AWS/Azure) · Database Optimization · DevOps & CI/CD · AI/GenAI Integration · Ownership & Accountability · Decision-Making Under Ambiguity · Cross-functional Collaboration